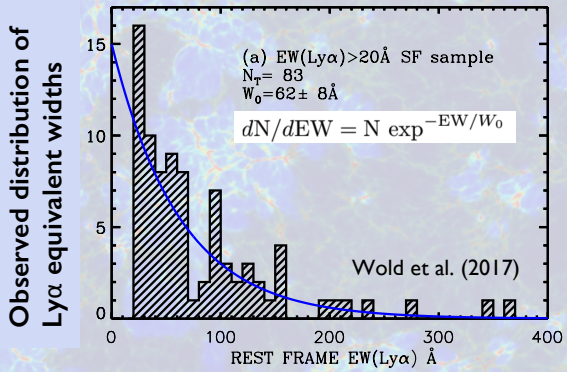
Constraining the End of Reionization with Deep Lyα Spectroscopy

Intae Jung[1], Steven Finkelstein[1], Rachael Livermore[2], Isak Wold[1], Mark Dickinson[3], Casey Papovich[4], Vithal Tilvi[5], and Rebecca Larson[1]

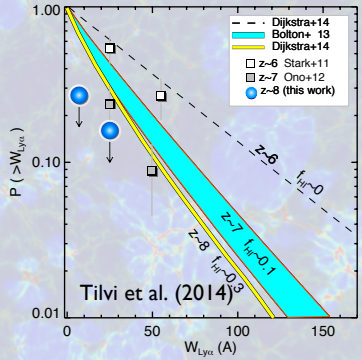
[1]The University of Texas at Austin
[2]The University of Melbourne
[3]National Optical Astronomy Observatory
[4]Texas A&M University
[5]Arizona State University

[1] Topic: Lyα emission as a probe of cosmic reionization

- Neutral hydrogens in the intergalactic medium affect the observed features of Lyα emission



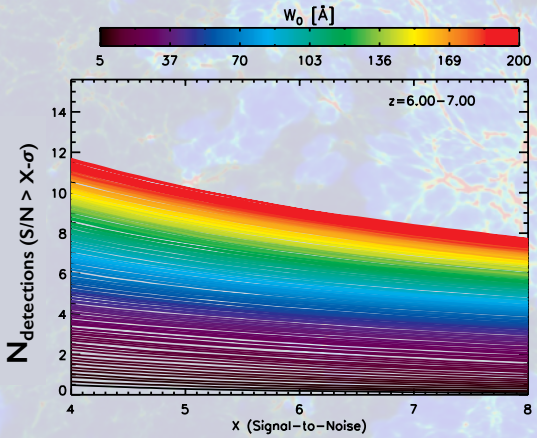
[Fig.1] Lyα galaxy EW distribution with the best fit exponential, well characterized by a scale length, $\mathbf{w_0}$.



[Fig.2] Observed Lyα EW distribution with the model predictions

[3] Method

Simulating the expected number of detections with varying Lyα equivalent widths distribution (W_0)

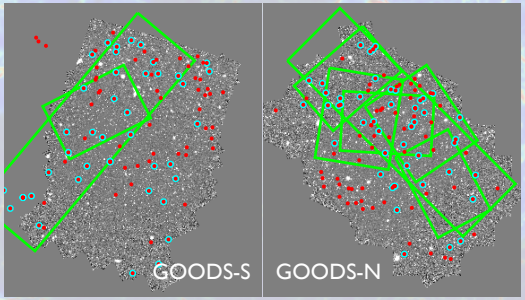


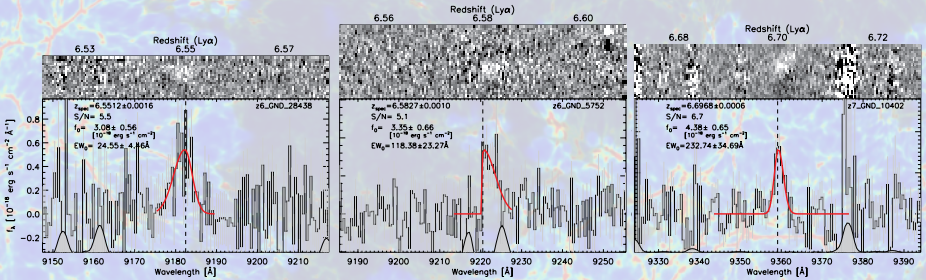
[Fig.5] A larger choice of W_0 predict more number of Lyα detections.

[2] Spectroscopic Observation for z = 5.5 – 8.2 galaxies

The DEIMOS (optical) and MOSFIRE (NIR) spectrographs on the Keck telescope

[Fig.3] The DEIMOS & MOSFIRE slitmask designs

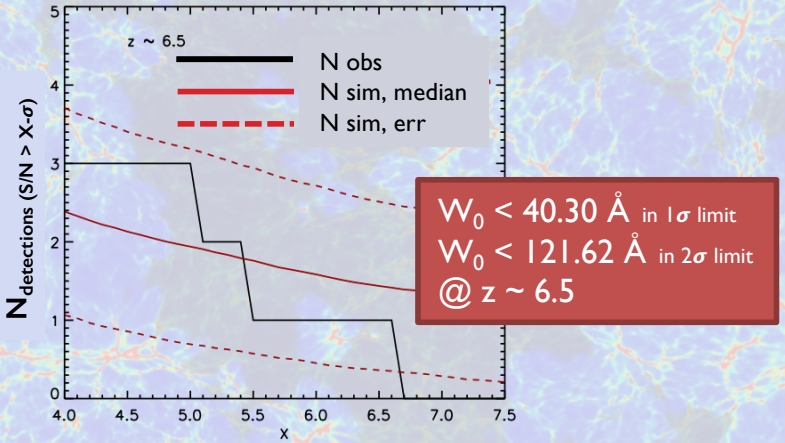




[Fig.4] Three detected Lyα emitters from the DEIMOS data

[4] Result

Constraining the distribution of Lyα equivalent widths



[Fig.6] Constrained probability of EW distribution with observed Lyα detections at z ~ 6.5.

Background image credit: Park, Shapiro et al. (2014)